

Lab: Neural Architecture for Active Inference

Learning Goal: Map cortical circuitry and subcortical systems onto Active Inference computations, analyzing empirical evidence.

Part 1: Cortical Hierarchy Mapping

Exercise: For each brain region, specify its level in the generative hierarchy, the type of predictions it generates, and the prediction errors it receives:

Brain Region	Hierarchy Level	Predictions Generated (top-down)	Prediction Errors Received (bottom-up)	Example
V1 (primary visual)	Lowest visual	Oriented edge expectations	Retinal contrast mismatches	Predicting edge at 45°
V4 (color/shape)	Intermediate visual	Shape and color expectations	V1 feature-combination errors	Predicting "round and red"
Inferior temporal (IT)	High visual	Object identity expectations	V4 shape errors	Predicting "apple"
Prefrontal cortex	Highest	Goal/context expectations	IT object relevance errors	Predicting "kitchen context"

Now trace a specific percept (e.g., recognizing a friend's face) through this hierarchy. What predictions flow down? What errors flow up? How is the percept "constructed"?

Write your response here

Part 2: Canonical Microcircuit Analysis

Learning Goal: Analyze how cortical layers implement predictive coding operations.

Exercise: For the Bastos et al. (2012) canonical microcircuit, explain the computational role of each layer and how damage to that layer would disrupt Active Inference:

Layer	Computation	Effect of Damage
L1 (apical dendrites)	Receives top-down context modulation	Loss of contextual modulation; perception becomes context-free
L2/3 (superficial pyramidal)	Computes and sends prediction errors upward	Loss of error signaling; higher levels receive no updates
L4 (stellate cells)	Receives feedforward driving input	Sensory input is disconnected from cortical processing
L5 (deep pyramidal, thick)	Generates motor output and deep predictions	Loss of action generation and deep generative structure
L6 (deep pyramidal, thin)	Sends predictions to L4 and L1 of lower area	Loss of top-down predictions; perception becomes purely bottom-up

Write a 200-word analysis: What would perception look like if only feedforward connections remained (all feedback removed)? What does this predict about the experience of patients with specific lesion patterns?

Write your response here

Part 3: Subcortical System Analysis

Learning Goal: Map subcortical structures to Active Inference computations.

Exercise: Complete the analysis for each subcortical structure:

Structure	Active Inference Function	Key Neural Circuit	Clinical Significance
Basal ganglia (striatum)	Policy evaluation — expected value of actions	Cortex→Striatum→Pallidum→Thalamus→Cortex	Parkinson's: impaired policy selection
Cerebellum	Forward models for motor prediction	Pontine nuclei→cerebellum→red nucleus/thalamus	Ataxia: prediction errors in motor control
Thalamus (pulvinar)	Precision gating of prediction errors	Reticular nucleus inhibition of relay cells	Neglect syndromes: precision mis-allocation
Hippocampus	Temporal inference and episodic memory	CA3 recurrent collaterals→CA1→entorhinal	Amnesia: inability to contextualize in time

For each structure, explain what would happen to Active Inference if it were removed.

Write your response here

Part 4: Neuromodulation and Precision

Learning Goal: Analyze how neuromodulators implement precision weighting.

Exercise: Design a simple thought experiment for each neuromodulator:

1. **Dopamine (too much / too little):** How does dopamine excess (e.g., schizophrenia) vs. deficiency (e.g., Parkinson's) affect policy selection and reward prediction?
2. **Norepinephrine (too high / too low):** How does NE excess (e.g., panic attack) vs. deficiency (e.g., inattention) affect precision of environmental volatility?
3. **Acetylcholine (too high / too low):** How does ACh excess vs. deficiency (e.g., Alzheimer's) affect sensory precision?
4. **Serotonin (too high / too low):** How does serotonin excess vs. deficiency affect temporal discounting?

Write your response here

Part 5: Empirical Evidence Evaluation

Learning Goal: Critically assess the evidence for neural Active Inference.

Exercise: Evaluate three key predictions of neural Active Inference against empirical evidence:

Prediction	Evidence For	Evidence Against	Verdict
Feedforward = errors, feedback = predictions	Laminar fMRI, electrophysiology, lesion studies	Some feedback connections seem to carry error signals	
Cortical columns = generative model nodes	Anatomical organization, computational modeling	Column boundaries are less clear than assumed	
Neuromodulators = precision weighting	Psychopharmacology data, computational modeling	Multiple functions of each neuromodulator	

Write a 300-word assessment: Is neural Active Inference well-supported, or is it still largely theoretical?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Anatomical mapping	Cortical hierarchy ↔ generative model
2	Circuit analysis	Canonical microcircuit
3	Systems analysis	Subcortical contributions
4	Clinical reasoning	Neuromodulation and precision
5	Evidence evaluation	Empirical support